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A local, generally covariant field theory of modified inertia

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Version 7 (2026-08-09). Changes from earlier versions are listed in §12.

The theory in one box

$$S = S_{\text{EH}}[g] + S_{\text{kh}}[g, T] + S_{\chi}[g, u, \chi] + S_{\text{m}}[g, T, \chi, x]$$

$$S_{\text{kh}} = \frac{1}{16\pi G} \int N \sqrt{h} \left[K_{ij} K^{ij} - \lambda K^2 + \xi R^{(3)} + \eta a_i a^i \right], \quad a_i = \partial_i \ln N, \quad n_{\mu} = -\frac{\partial_{\mu} T}{\sqrt{-(\partial T)^2}}$$

$$(u^{\mu} \partial_{\mu} + m)^2 \chi = \frac{g|a|}{c}, \quad S_{\text{m}} = -mc^2 \int \left[\mu(\chi) d\tau + (1 - \mu(\chi)) dt \right]$$

Three things are true of this action that were not true of the worldline construction it completes [1]:

1. **It is local.** The memory kernel is gone — replaced by an auxiliary field χ whose retarded Green's function *is* that kernel (§2).
2. **It is generally covariant.** The preferred frame is the normalised gradient of a scalar, so the Lorentz violation is spontaneous rather than stipulated (§3).
3. **Its propagating content is $2 + 1 + 0 = 3$ modes, and all three are healthy** in an explicit nonempty window (§4), **with a strong-coupling scale that clears every scale the theory is applied at by dozens of orders, and no Vainshtein-type screening anywhere** (§5).

And the acceleration scale is no longer the first moment of a postulated kernel:

$$a_0 = \frac{2}{3} c \frac{m^2}{g}$$

the ratio of the auxiliary field’s mass squared to its coupling. Together with the companion no-go result [2] — that this moment is logarithmically divergent, so it cannot be computed from a free-field correlator — this means g/m^2 is a *renormalised coupling*. Which is what every coupling in every local field theory is.

The honest caption, which must travel with the box. a_0 ’s *value* is still not derived, and the covariantisation *added* two free parameters (λ, η) rather than removing any. The theory is complete in the sense that a local, covariant, ghost-free action exists, has a strong-coupling scale far above every regime it is applied in, and yields MOND; it is not complete in the sense of predicting its own constants. It says nothing whatever about particle physics (§11).

And one thing more, added in v6, and it is the most serious limitation in this paper. Taken as a *fundamental* theory the construction predicts that gravitational lensing traces the **baryonic** mass alone — the metric is unmodified and photons have no rest mass to modify — which is excluded by the observed agreement between lensing and dynamical masses at $\sim 21\sigma$ (§8). **So what is established below is a complete and healthy field theory of the *test-particle sector*. Its standing as the fundamental theory is not established, and §8 states what would have to replace it.**

1. What was missing

Reference [1] gives a worldline action reproducing Milgrom’s modified-inertia relation via the *rapidity gap*, $\cosh \theta = -u \cdot u' / c^2$, with the memory encoded in a kernel $K(s)$ whose first moment satisfies $M_1 = \frac{2}{3} c / a_0$. Three things were owed:

- **Nonlocality.** $\Theta(\tau) = \int_0^\infty ds K(s) \theta(\tau, \tau - s)$ has no local derivative expansion, so there is no effective-field-theory power counting and no Hamiltonian.
- **Ghosts.** The equation of motion is fourth order in x . Ostrogradsky’s theorem does not apply, because it requires a *local* higher-derivative Lagrangian — but that is a statement about the theorem’s silence, not a proof of ghost-freedom.
- **A stipulated frame.** The action uses dt , the cosmological time, and a fixed timelike n^μ . That breaks general covariance by hand, and it makes the paper’s Lorentz-violation prediction dismissible as an artefact of choosing n .

This paper discharges all three. Each section’s claims are produced by a committed script that exits non-zero on any internal failure and carries negative controls that must trip (§12).

2. Step 1: the kernel is a local field

The reduction. For a general worldline the rapidity gap obeys $\theta(\tau, \tau - s) = (s/c) |a(\tau - s/2)| + O(s^3)$ [1] — the gap across an interval is s/c times the acceleration magnitude at the **midpoint**. So Θ carries both a factor of s and a lag of $s/2$. Substituting $s = 2u$ collapses *both* into one kernel:

$$\Theta(\tau) = \int_0^\infty du G(u) J(\tau - u), \quad G(u) = 4u K(2u), \quad J = |a|/c.$$

For the minimal causal kernel $K(s) = (N/\lambda)e^{-s/\lambda}$ this is

$$G(u) = g u e^{-mu}, \quad m = \frac{2}{\lambda}, \quad g = \frac{4N}{\lambda},$$

whose zero-frequency gain is $\int_0^\infty G du = g/m^2 = N\lambda = M_1$ **exactly**: the kernel’s first moment *is* the local operator’s DC gain.

Theorem 1. $g u e^{-mu}$ is the retarded Green’s function of $(d/d\tau + m)^2$.

Verified three independent ways: it solves the homogeneous equation for $u > 0$; its jump conditions are $G(0) = 0$, $G'(0) = g$, so $(d/du + m)^2 G = g \delta(u)$; and its Laplace transform is $g/(p + m)^2$. Hence

$$\ddot{\chi} + 2m\dot{\chi} + m^2\chi = \frac{g|a|}{c}, \quad \Theta = \chi$$

— **a damped oscillator whose friction term is the memory.** This is closed numerically as well as formally: integrating the local ODE forward with retarded data reproduces the nonlocal convolution to 10^{-23} , and perturbing m by 5% breaks that agreement by 5×10^{-2} .

The damping is critical, exactly, and it is forced. The discriminant of $p^2 + 2mp + m^2$ vanishes identically; there is a double root at $p = -m$. And for $K \sim s^k e^{-s/\lambda}$ the local operator has order $k + 2$, so the exponential kernel gives the *minimal* localisation. The extra factor of u traces to the rapidity gap being **linear in s** — which is the same fact that, in the companion paper [2], places the required kernel moment exactly on the pole of ζ . One structural feature, three consequences.

The ghost question becomes a computation. Enforcing the constraint with a multiplier π , the adjoint of $(d/d\tau + m)^2$ is $(d/d\tau - m)^2$ — *anti*-damped, the Bateman mirror — and the (Θ, π) kinetic form is off-diagonal, with eigenvalues $\pm \frac{1}{2}$ and signature $(+, -)$. By the naive criterion, a ghost. **It is not a new degree of freedom.** π is a *costate*: it carries a final-value condition rather than Cauchy data, it decays going backward ($|\pi| \sim e^{-m(T-\tau)}$, bounded by 1 on the physical interval), and the count closes exactly — 2 conditions on Θ plus 2 on $\pi = 4$, which is the order of the fourth-order equation of motion in x already derived in [1]. The indefinite metric is the Keldysh $(-)$ branch of the in-in formalism the construction already uses. This *replaces* the “Ostrogradsky is silent” argument with a positive statement.

a_0 **becomes a coupling ratio.** From $a_0 = \frac{2}{3}c/M_1$ and $M_1 = g/m^2$:

footing	M_1 (Gyr)	λ (yr)	m (s ⁻¹)	g (s ⁻¹)	N	a_0 rebuilt
canonical	67.65	3.2×10^4	1.9805×10^{-12}	8.374×10^{-6}	2.114×10^6	9.3619×10^{-11}
ALT ($\times 1.2048$)	56.15	3.2×10^4	1.9805×10^{-12}	6.951×10^{-6}	1.755×10^6	1.1279×10^{-10}

reproducing a_0 to 10^{-25} on both footings, with the implied kernel weight matching the independently derived bound $N \geq 2.1 \times 10^6$ of [1].

Costs, in the same breath. (i) The localisation is exact for the *midpoint* form and only third-order accurate for the exact bilocal action; galactically $\lambda\Omega = 8.8 \times 10^{-4}$, so the relative error is $O((\lambda\Omega)^2) = 7.7 \times 10^{-7}$ — small, not zero. (ii) In x the localised system is *second*-derivative, because the source is $|a|$, so **Ostrogradsky is not evaded in the localised writing**. The exact action contains only u , so this is an artefact of the midpoint expansion — which is precisely why “Ostrogradsky-free” and “fourth-order equation of motion” coexisted in [1] rather than contradicting each other. (iii) **Exact localisation of the full bilocal $\cosh^{-1}(-u \cdot u'/c^2)$ is not available by this route at all:** the auxiliary-propagator trick requires linearity in the delayed field. A genuine limitation.

Covariantly, $(u^\mu \partial_\mu + m)^2 \Theta = g|a|/c$ is a scalar advected by the matter flow obeying second-order relaxation — structurally the Israel–Stewart class [3], whose purpose was to restore the causality and hyperbolicity that first-order dissipative hydrodynamics destroys. A route and an analogy; no Israel–Stewart result is imported.

3. Step 2: the preferred frame becomes dynamical

Replace n^μ by the normalised gradient of a scalar khronon, $n_\mu = -\partial_\mu T / \sqrt{-(\partial T)^2}$. It is unit timelike by construction, and invariant under the reparametrisation $T \rightarrow f(T)$: carrying $f'(T)$ as a positive symbol, the derivative of n with respect to it vanishes identically. So the **foliation is physical and the labelling is gauge**.

Theorem 2 (the vorticity vanishes identically). *For any n built from a gradient, $\partial_{[\mu} n_{\nu]} = A_{[\mu} n_{\nu]}$ with $A = -\partial \ln N$, and the spatial projector annihilates n ; hence $\omega_{\mu\nu} = 0$ in every metric.*

Verified termwise on a generic T and generic N — T ’s second derivatives cancel — and the Christoffels drop out of the antisymmetrisation, so the result is metric-independent. **Consequence: the realisation is the hypersurface-orthogonal (Hořava) sub-case of Einstein-aether [4,5,6], not the general aether, and the spin-1 sector is removed entirely.** A prespecified non-gradient decoy field has nonzero curl, confirming the theorem is about gradients and not about the antisymmetrisation.

Theorem 3 (the khronon’s acceleration is the Newtonian field).

$$a^\mu[n] = \partial^\mu \Phi \quad (\text{linear order, static weak field})$$

Computed from the Christoffels of an explicit metric, not quoted; it is purely spatial ($a^t = 0$) and equals the log-lapse gradient, as the general $K/\omega/a$ decomposition requires. Three prespecified decoys — $2\partial\Phi$, $-\partial\Phi$, $\partial\Phi/2$ — are all rejected, so this measures the coefficient *and* the sign.

This matters more than the repair does. Before covariantisation, the only acceleration in the theory was the particle’s own $|a|$. MOND is a relation between g_{obs} and g_{bar} , and g_{bar} now exists as a *geometric object*. §6 draws the consequence.

Nothing is lost. In the gauge $\sqrt{-(\partial T)^2} = 1$ with $T = t$, the covariant coupling $\sqrt{(u \cdot n)^2}$ equals γ **exactly** — so the action of [1] is the *unitary gauge* of the khronon action. The covariantisation

is a completion, not a modification. It remains CPT-even for the same reason Form III of [1] was: $(u \cdot n)^2$ is invariant under $n \rightarrow -n$ while $(u \cdot n)$ flips.

The couplings collapse from four to three. With $\omega = 0$ the decomposition $\nabla_\mu n_\nu = K_{\mu\nu} - n_\mu a_\nu$ gives $(\nabla_\mu n_\nu)(\nabla^\mu n^\nu) = K \cdot K - a \cdot a$ and $(\nabla_\mu n_\nu)(\nabla^\nu n^\mu) = K \cdot K$, so c_1 and c_3 differ only by the acceleration term, which is degenerate with c_4 :

$$\sum_i c_i T_i = (c_1 + c_3) K \cdot K + (c_4 - c_1) a \cdot a + c_2 (\text{tr } K)^2$$

with Jacobian rank 3 — matching the three parameters of the infrared limit of non-projectable Hořava gravity [4]. Reinstating a vorticity term makes T_1 and T_3 independent again with coefficient $(c_1 - c_3)$, confirming the collapse is a consequence of Theorem 2 and not an algebraic accident.

Propagating content: 2 tensor + 1 scalar, no spin-1.

And the Lorentz-violation prediction becomes a real coupling. $|B| = (1 - \mu)/2 \approx a_0^2/8g^2$ [1] previously multiplied a stipulated n ; with n dynamical it is the coefficient of a matter-khronon coupling of CPT-even $c^{\mu\nu} \sim B n^\mu n^\nu$ type, so the g^{-2} scaling can no longer be dismissed as an artefact of the choice of frame.

4. The spin-0 health check

This is the calculation that could have killed §3, so it is done from scratch rather than by quoting a dispersion relation. Perturbing in the scalar sector, $N = 1 + \alpha$, $N_i = \partial_i B$, $h_{ij} = (1 + 2\zeta)\delta_{ij}$, and working in Fourier so every integration by parts is algebra:

$$S_2 = \int dt \left[3(1-3\lambda)\dot{\zeta}^2 + 2(1-3\lambda)k^2\dot{\zeta}B + (1-\lambda)k^4 B^2 + 2\xi k^2 \zeta^2 + 4\xi k^2 \alpha \zeta + \eta k^2 \alpha^2 \right]$$

with α and B carrying no time derivative — they are constraints.

The GR check passes first. The α constraint gives $\alpha = -2\xi\zeta/\eta$, and **at $\eta = 0$ it degenerates to $4\xi k^2 \zeta = 0$, forcing $\zeta = 0$: general relativity has no propagating scalar.** Reproduced, not assumed — so the mode found below is genuinely the Lorentz-violating one. Eliminating both constraints leaves $A = 2(1-3\lambda)/(1-\lambda)$ and

$$c_s^2 = \frac{\xi(2\xi - \eta)(1 - \lambda)}{\eta(1 - 3\lambda)} \stackrel{\xi=1}{=} \frac{(2 - \eta)(\lambda - 1)}{\eta(3\lambda - 1)}$$

The tensor sector independently gives $\frac{1}{4}\dot{\gamma}^2 - \frac{\xi}{4}k^2\gamma^2$: graviton speed² = ξ with a *positive* kinetic term, which is what anchors the overall sign convention — and means **GW170817 makes $\xi = 1$ a measurement to $\sim 10^{-15}$, not a choice.**

The window, and the two diseases are one band:

- **No ghost** $\iff A > 0 \iff \lambda > 1$ or $\lambda < 1/3$.
- **No gradient instability** $\iff c_s^2 > 0 \iff 0 < \eta < 2$.

Both exclude **exactly** $1/3 < \lambda < 1$, where the mode is simultaneously a ghost and gradient-unstable. The healthy region is nonempty and explicit:

(λ, η)	$(2, 1)$	$(2, \frac{1}{2})$	$(1.0001, 10^{-4})$	$(0.2, 1)$	$(0.2, \frac{1}{2})$
c_s^2	0.20	0.60	0.9998	2.0	6.0

Four prespecified sick decoys are rejected; rescaling ζ leaves c_s^2 unchanged, so it is a physical speed; and flipping the action’s overall sign makes the *graviton* go bad, confirming the no-ghost criterion is anchored to the tensor sector.

And observation points the right way. Preferred-frame PPN pushes both $(\lambda - 1)$ and η small, and in that corner

$$c_s^2 \rightarrow \frac{\lambda - 1}{\eta} \quad \text{exactly}$$

— **only the ratio survives**, so smallness alone drives c_s neither to zero nor to infinity. The vacuum Cherenkov bound [7] requires the scalar to be at least luminal, since a subluminal mode would be radiated by ultra-high-energy cosmic rays. So it reduces to a single inequality,

$$\lambda - 1 \geq \eta > 0,$$

and superluminal propagation is *safe* here precisely because there is a preferred frame. **PPN and Cherenkov make independent demands: no conflict.**

Against interest. (i) **The health is achieved by choice, not predicted.** §3 bought general covariance at the price of two new free parameters, and nothing in modified inertia forces either; “the scalar is healthy” means “a viable region exists”, which is much weaker than a prediction. The theory *gained* parameters. (ii) **The $\eta \rightarrow 0$ corner that PPN prefers is the known strong-coupling corner** ($c_s^2 \rightarrow \infty$ at fixed λ): the strong-coupling scale of the infrared non-projectable theory falls as the Lorentz-violating couplings do [5], so PPN safety pushes the cutoff *down*. **That scale is not computed here**, and it is the sharpest remaining worry about the covariantisation. (iii) The α_1, α_2 formulas are not derived; only their qualitative pressure is used, and the 10^{-7} figures are standard solar-system orders [8].

5. Strong coupling and the static nonlinearity

5.1 The strong-coupling scale

Section 4 leaves one danger: preferred-frame PPN pushes $(\lambda - 1)$ and η small, and a kinetic term proportional to a small number is exactly when self-interactions turn on early. That was the sharpest open risk in v1 of this paper, and it is now computed.

Why the danger is real, stated precisely. Restore the khronon fluctuation, $T = t + \pi$. Then

$$\ln N = -\dot{\pi} + \frac{1}{2}\dot{\pi}^2 + \frac{1}{2}(\partial\pi)^2 + O(\pi^3), \quad a_i = -\partial_i\dot{\pi} + O(\pi^2), \quad K_{ij} = -\partial_i\partial_j\pi + O(\pi^2),$$

and **at** $\lambda = \xi = 1$, $\eta = 0$ **the khronon action vanishes identically** — the K sector gives $K \cdot K - K^2 = k^4 \pi^2 - k^4 \pi^2 = 0$, so π is pure gauge in general relativity, as it must be. A prespecified $\lambda = 3/2$ decoy does *not* cancel, so this detects the GR limit rather than an artefact of the substitution. Everything that survives is therefore proportional to the small parameters:

$$S_2 \sim M_{\text{Pl}}^2 \left[-\delta (\partial^2 \pi)^2 + \eta (\partial_i \dot{\pi})^2 \right], \quad \delta = \lambda - 1.$$

And an independent cross-check falls out. That action's dispersion is $\eta \omega^2 k^2 = \delta k^4$, i.e.

$$c_s^2 = \frac{\delta}{\eta} = \frac{\lambda - 1}{\eta}$$

which is **exactly** the PPN-corner limit of §4 — obtained there from the unitary-gauge ζ with the ADM constraints eliminated, and here from the Stückelberg π in flat space. Two independent gauges and variables, one answer; a prespecified decoy $\delta/(2\eta)$ is rejected.

The static nonlinearity of the η sector vanishes. Canonical normalisation from the η term is $\pi_c = \sqrt{\eta} M_{\text{Pl}} k \pi$, and the leading self-interaction is

$$-2\dot{\pi}(\partial_i \dot{\pi})^2 - 2\partial_i \dot{\pi} \partial_j \pi \partial_i \partial_j \pi$$

— **every term carries a time derivative, so the whole cubic part vanishes for static configurations.** There is no static Vainshtein-type screening radius from this sector; the nonlinearity needs time dependence to switch on. A decoy cubic $(\partial_x \pi)^3$ does not vanish statically, so this is a property of the actual terms. Derivative counting on the same terms gives cubic/quadratic $= c_s E / (\sqrt{\eta} M_{\text{Pl}})$, hence

$$\Lambda_{\text{sc}} \sim \frac{\sqrt{\eta} M_{\text{Pl}}}{c_s} \approx 7.7 \times 10^{14} \text{ GeV} \quad (\eta = 10^{-7})$$

So the PPN-preferred corner does lower the cutoff, but only as $\eta^{1/2}$, and from the Planck mass.

The comparison that settles it, built to be robust to the power. The theory is applied at absurdly low energies. Scanning $\Lambda_{\text{sc}} = \eta^p M_{\text{Pl}}$ over $p = \frac{1}{2}, 1, 2, 3, 4$ — deliberately allowing powers far worse than the one derived:

scale where the theory is applied	energy	Λ_{sc}/E at $p = \frac{1}{2}$	at $p = 4$
Milky Way orbital frequency	$5.7 \times 10^{-31} \text{ eV}$	1.4×10^{54}	4.3×10^{29}
Milky Way inverse size (8.2 kpc)	$7.8 \times 10^{-28} \text{ eV}$	9.9×10^{50}	3.1×10^{26}
solar system, 1/AU	$1.3 \times 10^{-18} \text{ eV}$	5.8×10^{41}	1.9×10^{17}
laboratory, 1/m	$2.0 \times 10^{-7} \text{ eV}$	3.9×10^{30}	1.2×10^6

Λ_{sc} exceeds every scale at which the theory is applied, for every power tested — 26 orders of galactic margin even at the pessimistic $p = 4$, and 50 at the derived $p = \frac{1}{2}$. So the strong-coupling scale bears on whether this is a UV-complete *quantum* theory, which it never claimed to be, and **not** on the phenomenology.

Against interest, and this corrects a claim of my own. A first draft of the accompanying script asserted that the cutoff clears every scale *including* the LHC (1.4×10^{13} eV). **That is false: at $p \geq 3$ the cutoff falls below collider energies**, so the khronon effective theory would not cover the LHC at those powers. It is harmless only because the matter–khronon coupling there is $|B| \sim a_0^2/8g^2 = 1.1 \times 10^{-23}$, and the *derived* power $p = \frac{1}{2}$ clears the LHC by 10 orders — but the claim needed correcting rather than softening. Two further caveats: **only the scaling is computed, not the coefficient** (§5’s table is built to be insensitive to it, but a factor of 100 in the prefactor is excluded by nothing here); and **the δ -sector’s static nonlinearity is not analysed** — the η sector’s static cubic dies, but the $(\partial^2\pi)^2$ sector’s cubics need not, and a Vainshtein radius from *that* sector is uncomputed. That is now the honest residual. The analysis is flat-space throughout.

A note on which theory this is. The notorious $\lambda \rightarrow 1$ strong coupling belongs to *projectable* Hořava gravity. The non-projectable “healthy extension” carrying the $a_i a^i$ term is the known repair [5] — and §3 landed on it **by theorem** rather than by choice, since the vorticity of a gradient-built n vanishes identically.

5.2 The static nonlinearity of both sectors

§5.1 established that the η sector’s cubic self-interaction vanishes for static configurations, and flagged the δ sector — the $-\delta K^2$ term — as unanalysed. That was v2’s sharpest remaining gap. It closes by a parity theorem.

Theorem 4 (K is odd in π , to all orders). *For a static khronon $T = t + \pi(\mathbf{x})$, the unit normal is $n^\mu = (1/w, -\partial_i \pi/w)$ with $w = \sqrt{1 - (\partial\pi)^2}$. Under $\pi \rightarrow -\pi$ the normalisation w is invariant — it depends on $(\partial\pi)^2$ — while $\partial_i \pi$ flips. Hence $K[-\pi] = -K[\pi]$ exactly.*

Verified on the full three-dimensional closed form with **no series truncation**. In one dimension the divergence has an exact closed form that makes the parity manifest:

$$K = -\frac{\pi''}{(1 - \pi'^2)^{3/2}}$$

— odd numerator, even bracket. Its expansion has **zero even orders** (checked through fifth): $-\pi''$, then $-\frac{3}{2}\pi''\pi'^2$, then quintic. And the general three-dimensional cubic $-\frac{1}{2}\partial^2\pi(\partial\pi)^2 - \partial_i\pi\partial_j\pi\partial_i\partial_j\pi$ reduces in one dimension to exactly $-\frac{3}{2}\pi''\pi'^2$, so the 3-D formula is *validated against* the exact 1-D result rather than asserted. Inserting a genuine even piece into n^i by hand gives K a nonzero quadratic term, so the theorem is a measurement and not an algebraic accident.

Corollary. K^2 is even, so the δ sector has no static cubic; its leading static self-interaction is quartic, with

$$\frac{\text{quartic}}{\text{quadratic}} = 3\pi'^2 \quad (\text{one dimension})$$

i.e. of *order* $(\partial\pi)^2$ with an $O(1)$ coefficient. (A first draft of the accompanying script wrote “exactly $(\partial\pi)^2$ ”; the coefficient is 3, and the script records the correction.)

And on the aligned static foliation it vanishes outright. For a static, shift-free metric the constant- t surfaces have $K_{ij} = \frac{1}{2N}(\dot{h}_{ij} - D_i N_j - D_j N_i) = 0$ identically, so both $K \cdot K$ and K^2 vanish at *every* order — only $\eta a_i a^i$ survives. And $\delta(K^2)/\delta T = 2K \delta K$ vanishes with K , which makes that foliation a **solution** of the K sector rather than an imposed ansatz. A time-dependent metric gives $K_{ij} \neq 0$, so this is a property of staticity and not of the formula.

The nonlinearity, priced. $|\partial\pi|$ is the tilt of the khronon foliation relative to the local frame, i.e. $\sim v/c$:

setting	v	$(\partial\pi)^2$
solar system vs the CMB frame	369.8 km/s	1.5×10^{-6}
galactic rotation	220 km/s	5.4×10^{-7}
cluster velocities	1000 km/s	1.1×10^{-5}
a relativistic probe	$0.1c$	1.0×10^{-2}

So there is no Vainshtein-type screening radius: the nonlinearity never reaches $O(1)$. It would require $|\partial\pi| \rightarrow 1$ — a foliation boosted at near-light speed relative to the local frame — which happens near a black-hole horizon and nowhere else.

Combining the two sectors. The η cubic dies by carrying a time derivative; the δ cubic dies by parity in π . Two different mechanisms, so **the leading static self-interaction of the whole khronon sector is quartic** — worth a factor of 811 over a surviving cubic (1.5×10^{-6} instead of 1.2×10^{-3}). The result is not cosmetic.

What remains, and it is narrower than what it replaces. (i) **The full T field equation around a real source is not solved.** The aligned foliation is shown consistent with the K sector and the nonlinearity is priced *given* $|\partial\pi| \sim v/c$, but $\pi(r)$ is not obtained, so a larger $|\partial\pi|$ than the kinematic estimate is not excluded. (ii) **$|\partial\pi| \rightarrow 1$ near a black-hole horizon is exactly where this analysis fails;** universal horizons in Lorentz-violating gravity [12] are a real known issue and are not addressed. (iii) The η sector’s quartic is not computed either — only that its cubic vanishes.

6. Step 3: MOND from the joint field equations

The metric sector stays Newtonian — and the constraint that made the khronon healthy is what guarantees it. Modified inertia *presupposes* an unmodified gravitational field: its entire content is a modified *response* to a Newtonian Φ . Khronon corrections to Φ are $O(\eta, \lambda - 1) \lesssim 10^{-7}$, while MOND phenomenology needs Φ_{bar} only to $\sim 1\%$. **Five orders of margin.** So §4’s constraint is not a cost — it is a consistency requirement the framework needed anyway. A decoy $\eta = 0.5$ would leave the metric wrong by 50%, so this is a real comparison.

The mass question, confronted. From $E = mc^2[1 + \mu(\gamma - 1)]$ [1]: the rest energy is mc^2 , independent of μ , while the kinetic term is $m\mu v^2/2$. Therefore

$$m_{\text{grav}} = m, \quad m_{\text{inert}} = m\mu, \quad \text{and they differ.}$$

That is not a defect to be explained away — **it is the definitional content of modified inertia**. And the weak equivalence principle survives: $m\mu a = mg_{\text{bar}}$ gives $\mu a = g_{\text{bar}}$ with no reference to mass or composition, so universality of free fall is exact. What is violated is the *equality* $m_i = m_g$. A decoy μ depending on the mass leaves m in the answer, so the cancellation is a real property of $\mu(\Theta)$. The consequence runs in the framework’s favour: a galaxy’s **gravitating** mass is its **baryonic** mass. Energy-momentum conservation for a time-nonlocal modified-inertia theory is cited [9], not proved here.

The chain closes. With the Newtonian metric established independently, the worldline equation $d(m\mu v)/dt = -m\nabla\Phi$ gives (the accompanying script demonstrates the limits on the $\alpha = 2$ form for definiteness; the in-force exponential kernel of §11 has the same two limits, so the conclusion is unchanged — see [13])

$$\mu(g_{\text{obs}}/a_0) g_{\text{obs}} = g_{\text{bar}} \implies g_{\text{obs}} = \sqrt{a_0 g_{\text{bar}}} \implies v^4 = GMa_0$$

exactly — flat rotation curves and the baryonic Tully–Fisher relation — with **no new parameter**, since $a_0 = \frac{2}{3}c^2/m^2/g$ is inherited from §2. A decoy deep limit $\mu \sim Y^2$ fails to reproduce $v^4 = GMa_0$, so this tests the interpolation’s deep power and not the algebra of substitution. The closure is not circular: the Newtonian metric came from §5’s first paragraph, not from the MOND relation.

The BTFR normalisation, both ways. The framework predicts $A_{\text{fw}} = 1/(Ga_0) = 80.5 M_{\odot}(\text{km/s})^{-4}$ against McGaugh’s fitted $A = 47$ [10] — a ratio of 1.712, i.e. **0.234 dex in mass**, or $v_c = 105.6 \text{ km/s}$ at $10^{10} M_{\odot}$ versus 120.8. Three things must be said together:

- **Against interest: the ALT footing fits better** (ratio 1.421 vs 1.712).
- **But it is not a test of a_0 .** M_{bar} scales with the stellar mass-to-light ratio, and $A = 47$ was obtained at $\Upsilon_{3.6} = 0.5$; the framework needs $\Upsilon_{3.6} = 0.856$ — high, yet inside the literature spread and close to the 0.70 that an independent RAR fit of this corpus prefers. The intercept is Υ -degenerate.
- **And the Υ -free estimator accepts the canonical value.** The gas-dominated a_0 -line box $[0.84, 1.36] \times 10^{-10}$ contains 9.3619×10^{-11} .

So the offset is a mass-to-light question, not an a_0 question: **neither a win nor a deficit**.

7. A fork the covariantisation exposes

This is the genuinely new physical content, and it is a consequence of Theorem 3. The covariant theory possesses **two** acceleration scalars: the particle’s own $|a|$, and the khronon’s $|a[n]| = g_{\text{bar}}$. So Θ may be sourced by either:

source of Θ	theory
$\Theta[a]$	pure modified inertia — the construction of [1]
$\Theta[a[n]]$	a third theory — external-field-driven; neither pure MI nor AQUAL

The second branch shares the same interpolation function and the same a_0 , and is a genuinely different theory. **The worldline formulation could not even express this fork**, because it had no covariant object equal to g_{bar} .

They are observationally distinguishable. In a directional external-field test — the aligned versus anti-aligned asymmetry of rotation curves relative to the external field direction — **pure modified inertia predicts exactly zero**, while an external-field-driven Θ does not. The fork is exposed here and **not resolved here**.

8. The lensing axis: what the construction cannot do

This is the sharpest negative result in the programme. It is a *theory* result — it follows from the construction’s own premise, not from any observer’s calibration — and it appears before the confrontation section because it changes what is being confronted.

The argument, in four steps.

1. **The metric is unmodified and sourced by baryons.** Not an assumption imposed to manufacture a problem: it is the premise §6 *verifies* to 10^{-7} . Modified inertia *needs* $\nabla^2\Phi = 4\pi G\rho_{\text{bar}}$, because its whole content is a modified response to a Newtonian field.
2. **Photons have no rest mass to modify.** The worldline action is exactly proportional to m — it vanishes identically at $m = 0$ — so light follows null geodesics of the baryonic metric.
3. Hence $M_{\text{lens}} = M_{\text{bar}}$, while a Newtonian reading of the anomalous *dynamics* gives $M_{\text{dyn}} = M_{\text{bar}}/f_{\text{bar}}$:

$$\frac{M_{\text{dyn}}}{M_{\text{lens}}} = \frac{1}{f_{\text{bar}}} = 6.4 \quad (\text{clusters})$$

4. **Observed:** 1.0–1.3. Lensing and X-ray/dynamical masses agree at the 10–30% level. On a deliberately generous 25% systematic the prediction is excluded at **21**, and still at 4.2σ if that systematic is inflated **fivefold**. Modified gravity predicts exactly 1, so the data do not merely disfavour this reading — they land on the alternative. It is a **ratio**, so much of the mass calibration cancels, and the lensing side assumes no equilibrium.

And the enhancement cannot sit in both places. The tempting repair is to modify the metric as well, so light sees the enhanced field. It fails arithmetically:

	equation of motion	result
modified gravity only	$a = \nu g_{\text{bar}}$	correct
modified inertia only	$\mu a = g_{\text{bar}}$	$a = \nu g_{\text{bar}}$, correct
both	$\mu a = \nu g_{\text{bar}}$	$a = \nu^2 g_{\text{bar}}$ — wrong by one factor of ν

The enhancement lives in the metric **or** in the inertia, never both. Modified inertia and modified gravity are *algebraically identical* for massive test particles — precisely why this framework could stay ambiguous about which it was — and **lensing is the observation that separates them. It selects the metric.**

Every escape was computed, and one closes less cleanly than a first draft claimed. The memory field χ cannot supply the missing lensing mass: it is a costate with no propagating mode, and the μ -independent part of the matter stress-energy is proportional to m , i.e. still baryonic. A photon coupling cannot be bolted on: $|B| = (1 - \mu)/2$ also enters multiplied by m and vanishes at $m = 0$, so giving light its own effective metric *is* the disformal/TeV $\bar{\text{S}}$ construction — modified gravity again, not a fifth option. A non-baryonic component works, but must be 84% of the cluster mass, at which point it lenses *and* supplies the dynamics and MOND is not needed in clusters at all: a replacement, not a rescue. **And the khronon is *not* intrinsically negligible** — that was the overstated claim. At $\eta = 1$ its effective density would be $159\times$ the cluster gas, and $\eta \approx 0.034$ would exactly supply the missing mass. **That escape is closed entirely by the preferred-frame PPN bound, a factor 3.4×10^5 , and by nothing else.** If that bound moved, the escape reopens.

What it costs. The lensing-viable arm of this framework is a modified-*gravity* realisation of the same kernel, the AeST/TeV $\bar{\text{S}}$ class [12,14]. **Then §§2–6 is not the fundamental theory: at best it is an effective description of test-particle dynamics inside a modified-gravity theory.** A demotion, stated as one.

It also exposes an inconsistency predating this paper. Using modified gravity for lensing and modified inertia for dynamics is not two descriptions of one theory — by the table above they are two *theories*, and the pair is inconsistent. **A consistency problem inside the framework, not a new observational tension.**

And three predictions are at stake, including the one §9 rests on. The external-field anisotropy (pure modified inertia predicts *exactly zero*), the g^{-2} Lorentz-violation scaling, and **the wide-binary γ_v target of the pre-registered test** are all modified-inertia signatures. If modified inertia is only an effective limit, these are predictions *of the limit* and must be **re-derived** in the modified-gravity theory rather than inherited. §9 should be read with that caveat attached.

What may be salvageable, not shown here. In AeST/TeV $\bar{\text{S}}$ -type theories the test-particle sector can carry an effective acceleration-dependent inertia, so the rapidity-gap construction may be recoverable as the **point-particle limit** of the modified-gravity theory rather than a rival to it. That would preserve the parity theorem (§5), the localisation (§6.1) and $a_0 = \frac{2}{3}cm^2/g$, while giving up the claim that inertia rather than gravity is what is modified. **Not demonstrated. It is the next calculation.**

9. Confrontation: the pre-registered wide-binary test

A field theory that cannot be killed is not worth much. This section states the live observational test, its registered numbers and — the part usually omitted — **its power**, computed before the data.

What this construction does and does not supply. It must be said plainly first: **the wide-binary prediction is not derived from steps 1–3.** It follows from the external-field effect applied to the framework’s interpolation kernel, both of which predate this paper. What §§2–6 supply is the *theory behind* the number — a local, covariant, ghost-free action in which that number is the prediction of something rather than of a rule. The number itself does not move.

The registered prediction. The observable is γ_v , the ratio of the measured velocity-scale to the

Newtonian one for wide binaries in the 2–30 kAU window:

$$\gamma_v = 1.1582, \quad \text{range } 1.1311\text{--}1.1964 \text{ (radial), } 1.1339\text{--}1.2007 \text{ (magnitude)}$$

on the exponential kernel of §11, over both a_0 footings and both external-field conventions. This is the author’s frozen, hash-stamped pre-registration as amended [13]; nothing in this paper alters it.

The one decisive statement, at the registered sample size $N = 30,000$. A Newtonian result across 2–30 kAU is evidence against the framework at **4.74–7.10 σ_{tot}** . That figure is clean for a specific reason: under a Newtonian truth the estimator’s shape bias is identically zero, so it survives every systematic the registration declares. **It is the number the theory should be judged on, and no re-hedging is permitted after the fact.**

Three things that limit the test, all registered in advance.

1. **The scoring bins collide with the prediction.** 1.1582 falls in the interval the frozen decision table pre-declared “MG-side; MI disfavored”, and the frozen modified-gravity target 1.137 lies *inside* the MI range, $0.77 \sigma_{\text{tot}}$ from the point value. A scorer executing the table on a measurement **at the framework’s own prediction** would record the framework as disfavored. The registration’s remedy is binding and is implemented: report the raw $\hat{\gamma}$ with σ_{fit} and **both** distances — to 1.000 and to 1.1582 — and never a single verdict word.
2. **A nuisance parameter sits outside its frozen window.** The estimator’s calibration nuisance lands at 1.0575–1.0959 against a registered window of $[0.95, 1.05]$, whose pre-declared consequence is “*systematic-limited, no verdict — reported, not repaired.*” The exponential kernel fails this on both variance treatments where the retired $\alpha = 2$ kernel passed both. This is a cost of the kernel change, and it is not repaired here.
3. **One corner is pre-declared unscorable.** On the magnitude convention the alternative-footing corner is 1.20069, which is 0.00069 **above** the > 1.20 no-verdict edge. A genuine detection there cannot be scored. Quoting only the radial convention would hide this at its convenient end.

Two further checks, both underpowered, and that is stated rather than discovered afterwards.

- **The anisotropy falsifier**, whose sign was pre-declared: perpendicular pairs must show the larger boost, and the opposite sense at $\geq 3\sigma$ falsifies the derived external-field effect *independently* of the aggregate. The 3-D eigenvalue spread is 0.176–0.199 — strengthened by the kernel change. **But only the sky-projected angle is observable.** Averaging $\gamma(\psi) = \gamma_{\perp} + (\gamma_{\parallel} - \gamma_{\perp}) \cos^2 \psi$ over isotropic 3-D orientations gives a projection dilution of $D = 0.2367$ by Monte Carlo over the registered $|b| > 15^\circ$ sky, against the in-plane closed form $4/(3\pi) = 0.4244$. So the **observable** split is 0.042–0.047, reaching only $1.00\text{--}1.13 \sigma$ at $N = 30,000$ and requiring $N \sim 2.1\text{--}2.7 \times 10^5$ for 3σ — **seven to nine times the registered sample.**
- **The gated branch**, which must be scored alongside the ungated one. In the aggregate it is $0.03\text{--}0.06 \sigma$ from Newton and would need $N \sim 8.6 \times 10^7$ to separate: **an aggregate-only scorer would report “Newtonian” for a universe in which the gated branch is true.** Its only handle is an internal rise across the window, 0.001σ at 2 kAU to 1.56σ at 30 kAU — falsifiable in *shape*, but 1.56 is not 3. A watch item. What the two *branches* separate by is large: the ungated signal is $135\times$ the gated amplitude.

And the fork of §7 has its own test. The choice of what sources Θ — the particle’s $|a|$ (pure modified inertia) or the khronon’s $|a[n]| = g_{\text{bar}}$ — is decided by a *directional* external-field measurement, in which **pure modified inertia predicts exactly zero** aligned asymmetry while an external-field-driven Θ does not. That is a separate front from the aggregate γ_v and it is not settled here.

Summary of the confrontation. One number with teeth (4.74–7.10 σ against a Newtonian result), three with context, one unscorable corner, and one nuisance outside its window. That is a weaker test than the registration originally expected — §1.5 of it anticipated a decidable outcome — and the honest position is that DR4 can *disfavor* this framework decisively and can *support* it only weakly.

10. Limitations

- **Single-stream: a formulation preference, not a restriction on the physics.** A first draft of this paper called this the sharpest gap. That was wrong, and the correction is recorded rather than absorbed. $\Theta(\tau)$ is an integral over *the particle’s own past*, so $\ddot{\chi} + 2m\dot{\chi} + m^2\chi = g|a|/c$ is an ODE in the particle’s own proper time: χ is a **per-worldline internal variable**, like a spin or an internal clock, and two stars crossing at a point simply carry different χ . Multi-stream is then a non-issue. And the metric sector never sees χ at all, because $m_{\text{grav}} = m$ is μ -independent (§6) — the first χ -dependent source term is the kinetic $m\mu v^2/2$, suppressed by $(v/c)^2 = 5.4 \times 10^{-7}$ at galactic speeds. So the single-stream restriction binds only a *continuum rewriting of χ undertaken for its own sake*: it does not restrict test-particle dynamics, which is what rotation curves are, and it does not touch the field equations.
- **The construction’s standing as a fundamental theory is not established** (§8). Taken literally it predicts baryonic lensing, excluded at $\sim 21\sigma$; the lensing-viable arm is a modified-gravity realisation, which makes §§2–6 an effective description of the test-particle sector. **This is the most serious limitation in the paper**, and it puts the three modified-inertia-specific predictions — including §9’s wide-binary target — in need of re-derivation.
- **Strong coupling and the static nonlinearity are now both computed** (§5) and neither threatens the phenomenology. Three narrower residuals remain in their place: only the *scaling* of Λ_{sc} is derived and not its coefficient; **the full T field equation around a real source is not solved**, so a larger $|\partial\pi|$ than the kinematic v/c estimate is not excluded; and the η sector’s quartic is uncomputed.
- **Strong field and near-horizon behaviour is the one regime where the analysis genuinely fails.** $|\partial\pi| \rightarrow 1$ invalidates §5.2’s expansion, and **universal horizons** in Lorentz-violating gravity [12] are not addressed. This is now the sharpest structural gap.
- **Nothing new is derived.** a_0 is the coupling ratio of §2, and μ ’s shape is an *input*: the framework’s in-force kernel is the exponential one, $\nu = 1/(1 - e^{-\sqrt{y}})$, adopted because **both** power-law kernels fail solar-system ephemerides — $\alpha = 1$ by $1279\times$ the Earth/Mars bound and $\alpha = 2$ by $8.5\text{--}12.4\times$ the Mars ranging budget, the latter because its $1/g$ tail binds at the *Sun* via the Jupiter reflex rather than at a planet [13]. **Steps 1–3 are insensitive to that choice**: the localisation concerns the *memory* kernel $K(s)$ and not $\mu(Y)$, so $a_0 = \frac{2}{3}c m^2/g$ carries no reference to μ ’s shape; and §6 used only the two limits $\nu \rightarrow 1/\sqrt{y}$ (deep) and $\nu \rightarrow 1$ (Newtonian), which the exponential kernel satisfies. Steps 1–3 establish *consistency and locality*, not predictive content.
- **The full bilocal cannot be localised** by the auxiliary-field route (§2).

- Strong fields, black-hole universal horizons, and nonlinear stability: not addressed. The analysis of §4 is flat-space, quadratic order, scalar sector.
- **Ostrogradsky is not evaded in the localised writing (§2)**, only in the exact one.

What a referee should attack first

1. **Whether Θ should be sourced by $|a|$ at all** — §6’s fork is a genuine ambiguity of the covariant theory, not a detail.
 2. **The near-horizon regime** — $|\partial\pi| \rightarrow 1$ breaks §5.2, and universal horizons are unaddressed.
 3. **The two new parameters λ, η** — whether a theory that gained parameters to become covariant has really gained ground.
-

11. What is not claimed

- a_0 ’s value is not derived, and $\kappa = \frac{1}{2}$ in $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ remains **fitted**. The companion no-go [2] shows *why* a free-field correlator cannot supply it.
 - **The theory acquired two parameters, not fewer.** General covariance and ghost-freedom were bought with λ and η , whose only constraints are the health window and PPN.
 - **No claim is made regarding particle physics, the Standard Model, or unification.** The author publicly withdrew earlier statements of that kind and does not restate them. A complete field theory of modified inertia is not a theory of everything, and nothing here is evidence for one.
 - **The interpolating function is not new.** $\nu = \sqrt{1 + 1/y}$ is Eq. (9) of Milgrom (1999).
 - The khronon sector is standard technology [4,5,6]; what is new here is its use to complete *this* construction, Theorem 3, and the fork of §6.
-

12. Reproducibility, and version history

Every quantitative claim above is produced by a committed script that exits non-zero if any internal consistency check fails, and each carries negative controls that must trip. All four are included:

- `mi_kernel_localisation_2026.py` — §2, 27/27 checks.
- `mi_khronon_covariantisation_2026.py` — §3, 26/26 checks.
- `mi_khronon_spin0_health_2026.py` — §4, 30/30 checks.
- `mi_khronon_strong_coupling_scale_2026.py` — §5.1, 24/24 checks. **New in v2.**
- `mi_khronon_delta_sector_static_2026.py` — §5.2, 22/22 checks. **New in v3.**
- `mi_step3_joint_field_equations_2026.py` — §§6–9, 32/32 checks.
- `mi_dr4_readiness_audit_2026.py` — §9, 31/31 checks. **New in v5.**
- `mi_dr4_anisotropy_and_gated_2026.py` — §9, 20/20 checks. **New in v5.**
- `mi_lensing_axis_2026.py` — §8, 24/24 checks. **New in v6.**
- `mi_cluster_measurement_audit_2026.py` — §8 context, 27/27 checks. **New in v6.**
- `mi_relativistic_completion_aest_2026.py` — §13.1, 28/28 checks. **New in v7.**
- `mi_dbi_khronon_2026.py` — §13.2, 28/28 checks. **New in v7.**
- `mi_dbi_cmb_class_run_2026.py` — §13.2, real CAMB+CLASS, 11/11 checks. **New in v7.**
- `mi_virialisation_verdict_2026.py` — §13.3, 21/21 checks. **New in v7.**
- `mi_shift_charge_ic_route_2026.py` — §13.4, 13/13 checks. **New in v7.**

- `mi_lyalpha_ic_route_confrontation_2026.py` — §13.4, 11/11 checks. **New in v7.**
- `mi_deser_levin_interpolation_2026.py` — §13.5, 24/24 checks. **New in v7.**
- `mi_graviton_bath_ctp_2026.py` — §13.5, 15/15 checks. **New in v7.**
- `mi_eps_tot_mode_counting_verdict_2026.py` — §13.5, 13/13 checks. **New in v7.**
- `mi_distance_free_gbar_estimator_sparc_2026.py` — §13.5, 14/14 checks, run on raw SPARC. **New in v7.**
- `mi_bulge_ml_cannot_be_pinned_2026.py` — §13.5, 10/10 checks. **New in v7.**
- `mi_third_category_search_2026.py` — §13 context, 39/39 checks. **New in v7.**
- `mi_condensate_vacuum_energy_a0_2026.py` — §13.5, 17/17 checks. **New in v7.**

Changes in v7

1. **New §13, the largest addition since v1.** The v6 demotion was taken seriously: the modified-gravity arm was completed as **AeST**, this paper’s kernel was shown to embed in it, and **§8’s 21.2σ lensing exclusion becomes 0.601σ with $\gamma_{\text{PPN}} = 1$.**
2. **A published no-go is dissolved, with a theorem explaining why.** A bounded (DBI-type) $K(Q)$ reverses Blanchet & Skordis 2024’s $455\times$ conflict, and for any power law $K \sim u^n$ the early-time $w \rightarrow 1/(n-1)$, so **only boundedness reaches $w = 0$** . Verified with a real CAMB+CLASS run.
3. **The interpolation function is derived exactly — and it is Milgrom 1999 eq. 9,** from the Deser–Levin construction. The paper now states that without hedging. **The same derivation forces $a_0 = 2cH_\Lambda$, excluded by SPARC at 15.6σ , so the de Sitter–Unruh heuristic must not be cited as support for $\kappa = \frac{1}{2}$.**
4. **κ is now reported as a MEASURED number, 0.551 ± 0.043 ,** via a distance-free estimator verified exactly immune on 175 raw SPARC galaxies (0.00 ppm across a 10% distance error). It is consistent with $\frac{1}{2}$ (1.20σ) **and** with $1/\sqrt{3}$ (0.61σ). The claim “ $\kappa = \frac{1}{2}$ exactly, unexplained” is replaced throughout.
5. **Four claims of the author’s own are withdrawn:** the “68% of Λ CDM dark matter in clusters” figure; an R^2 -lever cluster mechanism that satisfies *neither* published μ^2 bound; “ $\Lambda_D = O(1)$ is natural” (at $\Lambda_D = 1$ the khronon peaks *at* recombination and CLASS gives a 325% TT deviation); and a graviton-bath normalisation that landed on $\kappa = \frac{1}{2}$ exactly but is structurally wrong ($h_{00} = 0$ in TT gauge). **A bug in the author’s cluster audit is also recorded:** its quoted R_{500} field is the cluster *core* field.
6. **Clusters are sourced from a primordial shift charge,** which resolves Mistele, McGaugh & Hossenfelder 2023’s $2500\times$ objection, and **Lyman- α becomes a test of the external-field-versus-pointwise fork** rather than a constraint on the framework.
7. **Amendment 9 to the frozen Gaia DR4 pre-registration was filed,** moving the target to $\gamma_v = 1.2139$ as *provisional* and re-deriving the > 1.20 no-verdict edge to > 1.26 from its own definition, with the cost filed as a new declared risk.
8. Thirteen scripts added. Nothing in §§2–7 is retracted; §8’s demotion stands and is now *resolved* by the completion rather than merely acknowledged.

Changes in v6

1. **New §8, and it is the most serious limitation this paper carries.** Taken as a *fundamental* theory the construction predicts $M_{\text{lens}} = M_{\text{bar}}$, hence $M_{\text{dyn}}/M_{\text{lens}} = 1/f_{\text{bar}} = 6.4$ in clusters, against an observed 1.0–1.3: **excluded at 21σ** , and still at 4.2σ on a fivefold-inflated systematic.

2. **And the enhancement cannot sit in both the metric and the inertia** — doing both gives $a = \nu^2 g_{\text{bar}}$, wrong by one factor of ν . Modified inertia and modified gravity are algebraically identical for massive test particles, and **lensing is what separates them**.
3. **Consequence, stated as a demotion:** §§2–6 is at best an effective description of the test-particle sector inside a modified-gravity theory, and the three modified-inertia-specific predictions — including §9’s wide-binary target — must be **re-derived** rather than inherited. The abstract-level caption now says so.
4. **A claim of the author’s own is corrected:** the khronon was said to be intrinsically negligible. It is not — $\eta \approx 0.034$ would supply the missing lensing mass, and only the preferred-frame PPN bound (a factor 3.4×10^5) closes that escape.
5. The limitations list is reordered with §8 first, and “what a referee should attack” now opens with it. Two scripts added (24/24, 27/27).
6. Nothing in §§2–7 is retracted. Every equation and every check stands; what changes is the claimed *status* of the construction.

Changes in v5

1. **New §8: the confrontation.** The paper now states the live pre-registered wide-binary test, its registered numbers, and — the part usually omitted — **its power, computed before the data**.
2. **Said first and plainly: the wide-binary prediction is NOT derived from steps 1–3.** It follows from the external-field effect applied to the framework’s kernel, both of which predate this paper. §§2–6 supply the theory *behind* the number; the number does not move.
3. **The honest headline is that the test is asymmetric:** one decisive number (a Newtonian result is evidence against at $4.74\text{--}7.10 \sigma_{\text{tot}}$), against a scoring-bin collision, a nuisance parameter outside its frozen window, an unscorable corner, and two further checks that are underpowered at the registered N (anisotropy $1.00\text{--}1.13 \sigma$ needing $7\text{--}9\times$ the sample; the gated branch $0.03\text{--}0.06 \sigma$ in the aggregate). **DR4 can disfavor this framework decisively and can support it only weakly**, and §8 says so.
4. Two verification scripts added (31/31, 20/20). Sections 8–10 renumbered to 9–11.
5. Nothing else changes. a_0 ’s value is still not derived and no claim is made about particle physics.

Changes in v4

1. **A factual correction, found by auditing this paper against the author’s own frozen pre-registration and made rather than left standing.** v1–v3 stated that μ ’s shape is “the $\alpha = 2$ interpolation that solar-system ephemerides force.” **That is false:** $\alpha = 2$ misses the Mars ranging budget by $8.5\text{--}12.4\times$, because its $1/g$ tail binds at the Sun via the Jupiter reflex rather than at a planet, and the framework’s in-force kernel is the exponential $\nu = 1/(1 - e^{-\sqrt{g}})$ [13]. §9 now says so.
2. **Nothing structural moves.** The localisation concerns the *memory* kernel $K(s)$, not $\mu(Y)$, so $a_0 = \frac{2}{3}c m^2/g$ is untouched; and §6 used only the deep and Newtonian limits, which the exponential kernel satisfies. §6 now names the kernel used in the demonstration and records that the substitution is harmless.
3. No other change. a_0 ’s value is still not derived and no claim is made about particle physics.

Changes in v3

1. **New §5.2: the static nonlinearity of both sectors**, which was v2’s sharpest named gap. It closes by a **parity theorem** — K is odd in π to all orders, so K^2 is even and the δ sector has **no static cubic**. Its leading static self-interaction is quartic.
2. On the aligned static foliation the K sector vanishes at every order, and it is a *solution* rather than an ansatz, since $\delta(K^2)/\delta T \propto K$.
3. The nonlinearity is priced at $(\partial\pi)^2 \sim (v/c)^2 \leq 1.1 \times 10^{-5}$ everywhere the theory is applied: **no Vainshtein-type screening radius exists**.
4. Combining with §5.1, **the leading static self-interaction of the whole khronon sector is quartic** — the two cubics vanish for two different reasons.
5. **A claim of the author’s own is corrected**: a draft wrote the quartic/quadratic ratio as “exactly $(\partial\pi)^2$ ”; the coefficient is 3.
6. The limitations list is updated again: the δ -sector item is discharged, and the **near-horizon regime with universal horizons** becomes the sharpest structural gap.
7. Nothing else changes. a_0 ’s value is still not derived and no claim is made about particle physics.

Changes in v2

1. **New §5: the strong-coupling scale**, which was v1’s sharpest named risk. It is computed and **does not threaten the phenomenology**: $\Lambda_{\text{sc}} \sim \sqrt{\eta} M_{\text{Pl}}/c_s$, and the conclusion survives a scan over powers η^p , $p = \frac{1}{2} \dots 4$.
2. **A cross-check gained, not assumed**: the Stückelberg π formulation reproduces $c_s^2 = (\lambda - 1)/\eta$, which §4 obtained from unitary-gauge ζ — two gauges, one answer.
3. **The η -sector static nonlinearity is shown to vanish**, so there is no static Vainshtein-type screening from it.
4. **A claim of the author’s own is corrected rather than softened**: a first draft asserted the cutoff clears *every* scale including the LHC. It does not — at $p \geq 3$ it falls below collider energies. Stated in §5.
5. The limitations list is updated: the strong-coupling item is discharged and replaced by the narrower and now-sharpest gap, **the δ -sector’s static nonlinearity**.
6. Nothing else changes. Every equation, result and caveat of v1 stands, including that a_0 ’s value is not derived and that no claim is made about particle physics.

Both a_0 footings (canonical ρ_{DE} with cH_Λ ; $\text{ALT} \times 1.2048$) are carried on every dimensionful number.

AI-assistance disclosure. Portions of the analysis, numerical verification and drafting were carried out with the assistance of a large language model (Anthropic Claude). The author directed the work, specified and reviewed every load-bearing calculation, and takes full responsibility including for any errors. No AI system satisfies the criteria for authorship and none is listed as an author. Several intermediate claims produced during the work were found to be incorrect and were withdrawn before this deposit — among them a past-directed sign for n_μ , an assumed rather than derived reparametrisation invariance, an asserted BTFR band that was replaced by the computed value, the overstated single-stream limitation withdrawn in §8, an assertion that the strong-coupling scale clears collider energies at every power (corrected in §5.1), a cubic-order counter that returned zero terms and thereby made a check pass vacuously (found and replaced), and a quartic/quadratic ratio written as exact when its coefficient is 3 (corrected in §5.2).

13. The completion, and the coefficient (new in v7)

Version 6 demoted §§2–6 to an effective description of the test-particle sector inside a modified-gravity theory. This section reports what happened when that demotion was taken seriously and the modified-gravity arm was completed, tested, and pushed until it broke. **Two results are positive and structural; four are withdrawals of the author’s own earlier claims; and the coefficient is now a *measured* number rather than an unexplained one.**

13.1 The relativistic completion is AeST, and this kernel embeds in it

Of TeVeS, BIMOND, Aether-Scalar-Tensor and pure Bekenstein–Milgrom, exactly one reproduces the CMB, so the choice is forced rather than preferred: **AeST** (Skordis & Złośnik 2021, PRL **127** 161302). The free function factorises, and this framework supplies its Y -sector:

$$\mathcal{F}(Y, Q) = \frac{a_0^2}{8\pi G} \mathcal{F}_Y\left(\frac{Y}{a_0^2}\right) + K(Q).$$

From this paper’s own parametric pair $\mu = 1 - e^{-u}$, $x = u^2/\mu$: the deep-MOND requirement $\mu \rightarrow x$ holds exactly, $\mu \rightarrow 1$ Newtonian, $x(u)$ is a bijection ($h(0) = 0$, $h' = e^{-u}(1 + u) > 0$), the free function is **convex**, and the Newtonian residual is $e^{-\sqrt{y}} = 3.6 \times 10^{-3457}$ at Earth’s orbit. **And lensing clears quantitatively**, which is what §8 said was owed: $\Phi = \Psi$ gives $\gamma_{\text{PPN}} = 1$ and $M_{\text{dyn}}/M_{\text{lens}} = 1$ exactly, turning §8’s 21.2σ into 0.601σ . The g^{-2} Lorentz-violation prediction, lost in pure Bekenstein–Milgrom, is **restored** by the aether.

13.2 A bounded $K(Q)$ dissolves a published no-go, and a theorem says why

Blanchet & Skordis 2024 (JCAP **11** 040, §4.3.1) rule out the quadratic $K(Q) = \mu^2(Q - 1)^2$ — the form this completion would naturally use — because cosmology forces $\mu^{-1} \lesssim 0.22$ kpc while MOND needs $\gtrsim 100$ kpc: a $455\times$ conflict with **disjoint** intervals. Replacing it with a **bounded (DBI-type)** function,

$$K(Q) = -M^4 + \mu^2 \Lambda_D^2 \left[1 - \sqrt{1 - u^2/\Lambda_D^2} \right], \quad u \equiv Q - Q_0,$$

reverses the constraint rather than relaxing it. K' diverges as $u \rightarrow \Lambda_D$ so the shift current can grow as a^{-3} at all early times, while K itself **saturates** — bounded pressure with diverging density gives $w \rightarrow 0$, so the khronon is **dust** at early times instead of stiff. Cosmology then becomes a *lower* bound on μ^{-1} , pointing the same way as MOND. At $\mu^{-1} = 100$ kpc the quadratic gives $w(a = 3 \times 10^{-5}) = 0.9997$, failing by $61\times$; the bounded form gives 7.3×10^{-5} , a $226\times$ margin, ghost-free ($K'' > 0$) and subluminal throughout.

Theorem. For any power law $K \sim u^n$ the early-time equation of state is $w \rightarrow 1/(n-1)$: quadratic 1, quartic $1/3$, sextic $1/5$ — all fail. **No finite power reaches $w = 0$; only boundedness does.** That is precisely why the published polynomial escape needs $K_3 \sim 10^5$, which its own authors call unnaturally large, while the bounded form needs no large number at all.

A real CMB run, not a consistency argument. With CAMB and CLASS: at $\Lambda_D \leq 10^{-2}$ the khronon’s sound speed at recombination is $c_s^2 = 2.9 \times 10^{-8}$ and the TT spectrum is indistinguishable from CDM (0.069%, with $P(k = 0.2)$ to 1.7%). **This corrected an error of the author’s:** $\Lambda_D = O(1)$ was called natural, but there the khronon’s w and c_s^2 peak at 0.207 and 0.250 *at*

recombination ($z \approx 1749$) and CLASS returns a 325% TT deviation. The CMB forces $\Lambda_D \lesssim 10^{-2}$. Transience is load-bearing: holding c_s^2 at its peak for all time destroys $P(k=0.2)$, so **the acoustic peaks are verified and the post-recombination growth history is not.**

13.3 Dark matter in this completion: no particle, and none in galaxies

AeST fits the CMB *because* its scalar is dust, so **a dark component exists at the full Ω_{dm}** — removing it moves H_3/H_1 by 54%, and no refit of $A_s, n_s, H_0, \omega_b, \tau$ absorbs that. What does *not* exist is a dark-matter **particle**: the component is the Q -sector of the same scalar whose offset supplies Λ ($w = -1$ exactly at the minimum) and whose Y -sector supplies MOND. One function, three jobs.

And it is **absent where rotation curves are measured**. AeST’s quasi-static equation carries a Helmholtz term, so $\rho_c = -\mu^2 \Phi / 4\pi G$ tracks the *potential*, not the baryon density: for a flat rotation curve ρ_c is asymptotically flat and $M_c \sim r^3$ — centrally **evacuated**, the opposite of NFW. Hence $\xi \propto R^2$ and the galaxy-to-cluster ratio is parameter-free, $\xi(10 \text{ kpc})/\xi(R_{500}) = 5.1 \times 10^{-5}$. Clusters can carry the required $\xi = 0.11\text{--}0.26$ while galaxies carry $\xi \sim 10^{-5}$, an RAR cost of 6×10^{-5} dex against a 0.034 dex intrinsic scatter. **The defensible statement is “no dark-matter particle, and none in galaxies” — not “no dark matter.”**

Two corrections belong here. First, **a cluster figure of the author’s is withdrawn**: “the kernel removes only $\sim 32\%$ of cluster dark matter, leaving 68%” combined $1/f_{\text{bar}}$ with a banked $\eta_{\text{req}} = 2.334$ additively, which is incoherent. Computed directly and self-consistently — the back-reaction that added mass raises y and *lowers* ν — the kernel removes **74–89%**, leaving 11–26%. Second, **a bug in the author’s own cluster audit**: its $g = 2.02 \times 10^{-9} \text{ m s}^{-2}$ is the field at $\approx 0.19 \text{ Mpc}$, the cluster *core*, not R_{500} ; at true R_{500} clusters sit at 0.33–0.58 a_0 , near or below a_0 , so that audit’s quasi-Newtonian premise is wrong.

13.4 Clusters from a primordial shift charge, and it resolves a published objection

The dust amount I_0 is the conserved Noether charge of the shift symmetry $\varphi \rightarrow \varphi + c$, hence an *integration constant* fixed by an initial condition. If the amount is an IC then so is its spatial distribution, and a conserved charge is carried rather than created — so clusters can be khronon-rich for the same reason they are baryon-rich. This removes the cluster job from μ , and that single change **resolves the objection in Mistele, McGaugh & Hossienfelder 2023 (A&A 676 A100)**, whose Table 1 needs $\mu^2 \gtrsim 1 \text{ Mpc}^{-2}$ for clusters but $\lesssim 0.001$ for galaxy weak lensing — disjoint, up to 2500 $\times$. With clusters sourced primordially only the weak-lensing bound survives, and $\mu^2 = 5.2 \times 10^{-8}$ satisfies it by 1.9×10^4 .

It also kills a mechanism proposed one day earlier by the author: the R^2 lever’s required $\mu^{-1} = 3.13 \text{ Mpc}$ gives $\mu^2 = 0.102$, which *violates* the weak-lensing bound by 102 $\times$ and falls 9.8 $\times$ short of the cluster bound — it satisfies **neither**.

Lyman- α does not exclude the IC route, and the naive objection fails structurally: the forest observable is the MOND-enhanced total $\nu(y)(M_b + M_k)$, not the khronon’s transfer function. On the pointwise branch the compensation is the right size — at $z = 3$, $R_c \approx 1 \text{ Mpc}$, $\delta \approx 20$ the forest sits at $\nu = 4.08$ against a required 4.00, a 1.8% match from independent inputs. On the external-field branch the forest is Newtonian ($\nu = 1.004$) and the route is excluded at 4 $\times$. **So Lyman- α now tests that fork**, separating the branches by 30 $\times$ the forest’s 10% systematic floor. The (R_c, δ) spread is 5.2 $\times$, so this is an order-of-magnitude pass, not a clearance.

13.5 The coefficient: what is derived, what is not, and what is measured

The interpolation function is derived, exactly. From the Deser–Levin temperature of an accelerated observer in de Sitter, $T \propto \sqrt{a^2 + (cH)^2}$, plus inertia responding to the **excess** over the ambient bath, plus a Newtonian-limit normalisation that is forced rather than chosen:

$$\mu(a) = \frac{\sqrt{a^2 + (cH)^2} - cH}{a} \implies g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + a_0 g_{\text{bar}}}, \quad \nu = \sqrt{1 + 1/y}.$$

Nothing fitted. **This is Milgrom 1999** (Phys. Lett. A **253** 273, eq. 9), from exactly this construction; the interpolation is not this framework’s contribution, and this paper now says so without hedging.

And the same derivation forces $a_0 = 2cH_\Lambda = 1.083 \times 10^{-9}$, which SPARC excludes at 15.6σ (deep-MOND $g \propto \sqrt{a_0}$, so $11.6\times$ is a 0.53 dex offset against 0.034 dex). The construction is **rigid**: the power is forced to $n = 1$ by the two MOND limits, the baseline to the ambient de Sitter temperature, and the normalisation to unity by the Newtonian limit. No variant yields cH/Z . **Therefore the de Sitter–Unruh heuristic must not be cited as support for $\kappa = \frac{1}{2}$: it motivates the form and gets the scale wrong by an order of magnitude.** The apparent “exactly $2Z$ ” discrepancy is a **tautology** — $(2cH)/(cH/Z) = 2Z$ identically for any Z , and Milgrom’s rival 2π gives 4π by the same algebra.

Mechanism routes for κ itself. Every route that ties a_0 to Λ using only Λ, G, c produces the same algebra with a convention-dependent residue, i.e. a **relabelling**: $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$ is algebraically identical to $\frac{1}{2}\sqrt{G\rho_\Lambda}$ and to $Z = 2\sqrt{8\pi/3}$. A graviton-bath calculation does give the right **form** — the de Sitter horizon entropy cancels the Planck suppression exactly, $S_{\text{dS}}GH^2 = \pi$ identically, leaving $\kappa^2 = 8\pi\epsilon_{\text{tot}}$ with ϵ_{tot} a pure number — but **not the number**: five defensible readings span $\kappa = 0.013$ – 2.047 , and the one landing on $\frac{1}{2}$ exactly is **structurally wrong**, because $X = h_{\mu\nu}u^\mu u^\nu = h_{00}$ vanishes in TT gauge for a static worldline and is $(v/c)^2$ -suppressed *and velocity-dependent* otherwise, which breaks universality.

So κ is reported here as a measured quantity. Using a **distance-free** estimator — $g_{\text{bar}} = GM/R^2 \propto F/\theta^2$ is exactly invariant under $D \rightarrow D(1 + \delta)$ while $g_{\text{obs}} \propto 1/D$ is not, so profiling over a free vertical offset removes the distance-scale systematic entirely (verified on 175 SPARC galaxies: 0.00 ppm across a 10% distance error, against 28.9% for the standard fit) — the stat-plus-distance budget falls $6.83\% \rightarrow 1.84\%$, and

$\kappa = 0.551 \pm 0.043$

1.20σ from $\frac{1}{2}$ and 0.61σ from $1/\sqrt{3}$: **consistent with both.** The residual error is mass-to-light dominated, and that is structural rather than technological. The bulge M/L can neither be measured (the free four-parameter fit runs away to $a_0 = 4.2\times$ canonical) nor eliminated: cutting to the 143 bulgeless galaxies makes Υ_{bul} irrelevant and returns $\kappa = 0.4996$, but that is a **choice of $\Upsilon_{\text{disk}} = 0.5$** — at the value this framework’s own RAR fit prefers, $\Upsilon_{\text{disk}} = 0.70$, the same estimator gives $\kappa = 1.108$. Only 2.4% of bulgeless points lie above the knee, and below it a_0 and the offset are exactly degenerate. Refusing the (model-dependent) bulge/disc decomposition makes it worse still, $5.73\times$ versus $1.09\times$.

The honest form of the central claim is therefore $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ with κ measured = 0.551 ± 0.043 , consistent with $\frac{1}{2}$ — not “ $\kappa = \frac{1}{2}$ exactly, unexplained.” The observational floor

is $\approx 3.9\%$ on a_0 (helium, HI self-absorption, CO-dark H_2), independent of sample size, distances or stellar populations, so this line caps at $\kappa \pm 0.020$ — enough to separate $\frac{1}{2}$ from $1/\sqrt{3}$ at $\approx 4\sigma$, never enough to establish a rational.

13.6 The assembled action, and the registration

The full action with the factorised $\mathcal{F}(Y, Q)$, a fourteen-row verified table each backed by a named committed script, an honest parameter count (**four dark-sector numbers against Λ CDM’s two** — not fewer, but Λ , dark matter and MOND from *one* function) and an eight-item not-claimed list is written out separately in `THE_COMPLETION.md`, deposited alongside.

Amendment 9 to the frozen Gaia DR4 pre-registration was filed on 2026-08-09, moving the operative arm from modified inertia to modified gravity and the wide-binary target from $\gamma_v = 1.1582$ to $\gamma_v = \sqrt{\nu(y_{\text{extN}})} = 1.2139$ (canonical) / 1.2592 (alt), **registered as provisional** pending the full nonlinear AQUAL-EFE solve. The > 1.20 no-verdict edge was **re-derived from its own stated definition** — “above every EFE-saturated target”, which the new target falsified — to > 1.26 , with the cost filed as a new declared risk: the band 1.20 – 1.26 becomes scoreable, and the DR3 dry run’s 1.205 sits in it, excluded on §1.6’s own grounds and not on the edge. All eight prior hash files were left unmodified; the new digest is recorded in `AMENDMENT9_HASH.txt`.

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[13] Amendment 8 to the author’s frozen Gaia DR4 pre-registration (2026-08-03), which retires both power-law kernels on solar-system grounds and adopts $\nu = 1/(1 - e^{-\sqrt{y}})$; see also the companion Bekenstein–Milgrom field theory for that kernel, whose ellipticity, convexity, ghost-freedom and subluminality are proved for the same interpolation.